

# AWS vs Azure vs GCP

## General Comparison

| Feature               | AWS (Amazon Web Services)                              | Azure (Microsoft Azure)                         | GCP (Google Cloud Platform)                       |
|-----------------------|--------------------------------------------------------|-------------------------------------------------|---------------------------------------------------|
| Company               | Amazon                                                 | Microsoft                                       | Google                                            |
| Launch Year           | 2006                                                   | 2010                                            | 2008                                              |
| Market Share          | Highest (Largest Cloud Provider)                       | Second Largest                                  | Third Largest                                     |
| Popularity            | Most widely used worldwide                             | Popular in enterprises using Microsoft products | Popular among startups and AI/Data companies      |
| Learning Resources    | Huge community and documentation                       | Good documentation                              | Good documentation but smaller community          |
| Number of Services    | 240+ Services                                          | 200+ Services                                   | 150+ Services                                     |
| Ease of Learning      | Moderate                                               | Easy for Microsoft users                        | Beginner-friendly                                 |
| Free Tier             | 12 Months + Always Free Services                       | Free Credits + Free Services                    | Generous Free Credits + Always Free Services      |
| Global Infrastructure | Largest number of regions and availability zones       | Large global presence                           | Strong global presence but fewer regions than AWS |
| Best For              | General Cloud Computing, Startups, Enterprises, DevOps | Organizations using Microsoft Ecosystem         | AI, ML, Data Analytics, Kubernetes                |

## Compute Services (Virtual Servers)

| Purpose              | AWS                 | Azure                          | GCP                     |
|----------------------|---------------------|--------------------------------|-------------------------|
| Virtual Machine      | EC2                 | Virtual Machines               | Compute Engine          |
| Auto Scaling         | Auto Scaling Groups | Virtual Machine Scale Sets     | Managed Instance Groups |
| Serverless Functions | Lambda              | Azure Functions                | Cloud Functions         |
| Container Service    | ECS / EKS           | AKS                            | GKE                     |
| Container Registry   | ECR                 | Azure Container Registry (ACR) | Artifact Registry       |

## Storage Services

| Purpose        | AWS        | Azure         | GCP                   |
|----------------|------------|---------------|-----------------------|
| Object Storage | S3         | Blob Storage  | Cloud Storage         |
| File Storage   | EFS        | Azure Files   | Filestore             |
| Block Storage  | EBS        | Managed Disks | Persistent Disk       |
| Backup Service | AWS Backup | Azure Backup  | Backup and DR Service |

## Database Services

| Database Type       | AWS         | Azure                 | GCP                  |
|---------------------|-------------|-----------------------|----------------------|
| Relational Database | RDS         | Azure SQL Database    | Cloud SQL            |
| NoSQL Database      | DynamoDB    | Cosmos DB             | Firestore / Bigtable |
| Data Warehouse      | Redshift    | Synapse Analytics     | BigQuery             |
| In-Memory Cache     | ElastiCache | Azure Cache for Redis | Memorystore          |

## Networking Services

| Purpose         | AWS                         | Azure                  | GCP                  |
|-----------------|-----------------------------|------------------------|----------------------|
| Virtual Network | VPC                         | Virtual Network (VNet) | VPC                  |
| Load Balancer   | Elastic Load Balancer (ELB) | Azure Load Balancer    | Cloud Load Balancing |
| DNS Service     | Route 53                    | Azure DNS              | Cloud DNS            |
| CDN             | CloudFront                  | Azure CDN              | Cloud CDN            |

## Security & Identity

| Purpose             | AWS             | Azure                        | GCP                     |
|---------------------|-----------------|------------------------------|-------------------------|
| Identity Management | IAM             | Microsoft Entra ID           | Cloud IAM               |
| Key Management      | KMS             | Key Vault                    | Cloud KMS               |
| Secrets Management  | Secrets Manager | Key Vault                    | Secret Manager          |
| Security Monitoring | GuardDuty       | Microsoft Defender for Cloud | Security Command Center |

## Monitoring & Logging

| Purpose    | AWS             | Azure                | GCP              |
|------------|-----------------|----------------------|------------------|
| Monitoring | CloudWatch      | Azure Monitor        | Cloud Monitoring |
| Logging    | CloudWatch Logs | Azure Log Analytics  | Cloud Logging    |
| Tracing    | X-Ray           | Application Insights | Cloud Trace      |

## Messaging & Event Services

| Purpose              | AWS | Azure       | GCP     |
|----------------------|-----|-------------|---------|
| Message Queue        | SQS | Service Bus | Pub/Sub |
| Notification Service | SNS | Event Grid  | Pub/Sub |

## DevOps & CI/CD

| Purpose                | AWS            | Azure                 | GCP                        |
|------------------------|----------------|-----------------------|----------------------------|
| CI/CD Service          | CodePipeline   | Azure DevOps          | Cloud Build                |
| Source Control         | CodeCommit*    | Azure Repos           | Cloud Source Repositories* |
| Artifact Storage       | CodeArtifact   | Azure Artifacts       | Artifact Registry          |
| Infrastructure as Code | CloudFormation | ARM Templates / Bicep | Deployment Manager         |

## AI & Machine Learning

| Feature     | AWS                    | Azure                            | GCP                       |
|-------------|------------------------|----------------------------------|---------------------------|
| AI Platform | SageMaker              | Azure AI Services                | Vertex AI                 |
| Strength    | Wide range of AI tools | Strong enterprise AI integration | Strongest AI/ML ecosystem |
| Popular For | ML Model Deployment    | Business AI Solutions            | Data Science, AI, ML      |

## Kubernetes Support

| Feature            | AWS      | Azure     | GCP                            |
|--------------------|----------|-----------|--------------------------------|
| Managed Kubernetes | EKS      | AKS       | GKE                            |
| Ease of Use        | Moderate | Easy      | Best Kubernetes Experience     |
| Developed By       | Amazon   | Microsoft | Google (Creator of Kubernetes) |

## Pricing

| Factor             | AWS                 | Azure               | GCP                     |
|--------------------|---------------------|---------------------|-------------------------|
| Pricing Model      | Pay as You Go       | Pay as You Go       | Pay as You Go           |
| Cheapest Overall   | Depends on Workload | Depends on Workload | Depends on Workload     |
| Pricing Complexity | High                | Medium              | Medium                  |
| Discounts          | Reserved Instances  | Reserved Capacity   | Sustained Use Discounts |

## Industry Preference

| Industry                  | Preferred Cloud |
|---------------------------|-----------------|
| Startups                  | AWS / GCP       |
| Large Enterprises         | Azure           |
| Government Projects       | AWS / Azure     |
| AI & Machine Learning     | GCP             |
| Data Analytics            | GCP             |
| E-Commerce                | AWS             |
| Microsoft-Based Companies | Azure           |
| Hybrid Cloud Environments | Azure           |